

Trustworthy, Controllable, Causal

Three adjectives in one title, and the vocabulary I already said I did not have

The page for this vista is a heading and the word *Stewing*. That is accurate. What exists is a title, a folder, a submission plan with real dates in it, nine literature sections written against my notes, and one sentence of mine that commissions this file before it commissions anything else.

“As you can see here, i am using terms loosely and interchangeably, because I do not understand the differences. Can we find the standard language about this topic?”

brainstorm, 9 August, the threat-modeling bullet

So this is not a summary of a project that already knows what it is saying. It is the vocabulary the project has to fix before it says anything, and it records the places where my own pages already use one of these words in more than one way. I am not tidying those away. Four of them are in Section 5, and one of the four is the reason the title says *Many*.

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1 The frame

1.1 Three adjectives, three demands

The title carries three adjectives and each is a separate demand with a separate literature and a separate way of going wrong.

Trustworthy is a claim about what an operator can rely on. It is not a claim that the model is accurate. The Next page already states the version I mean, and it states it as a preference between two ways of being wrong.

“preferring a model that fails honestly (wide and uncertain) over one that fails silently (sharp and wrong)”

website, Cyber World Modeling, Next

Controllable is doing two jobs at once and I should stop pretending otherwise. Inside FOE-Dreamer it names a partition of the latent state, the block whose evolution is conditioned on the defender’s own action. In the title it names a property of the deployed system, that a person can constrain what the defender does and can say afterwards why it did it. Both senses are live in my own files and neither is going away.

Causal is the strongest of the three and the only one with an identifiability theory standing behind it. It is also the one my own Learn Structure page lists as a contrast case, in the sentence that says what strategic dependence is not. Four senses of the word are already in circulation across my pages, and they are catalogued below.

1.2 The claim the submission plan commits to

The plan in the vista folder is not vague about what the project asserts. It states a thesis, its null, and the experiments that separate them, which is the only reason this vista is a project rather than a mood.

“The failures of a world-model network defender are structured: they concentrate where the environment carries dependence types the world model cannot represent or learn, and they are therefore predictable in advance from a structural analysis of the environment”

submission plan, Claim

“Failure severity is explained by model bias alone, open-loop error growth over horizon, with no additional dependence on the type of structure in the environment”

submission plan, the null

Three experiments separate them. E1 trains the learner on generator instances that differ only in planted dependence type and degree and measures failure per type. E2 reads model bias directly off open-loop prediction error at horizons 1, 5 and 10, which is the null’s only explanatory variable. E3 asks whether E1’s type profile predicts which of the enumerated failure modes actually appear on the testbed at matched bias. The thesis predicts a type-ordered failure profile. The null predicts a flat one. Either result is publishable, which is the property that makes the plan worth keeping.

The motivating evidence is two things and both are small. In the generator, a structure learner

recovered all planted dependences but erred non-uniformly by type, with hallucinations concentrated in causal, three of five over ten seeds, and degree misestimation occurring only for causal. And my advisor’s stated expectation is that object-oriented world models learn spatial and control dependence, partially temporal, doubtfully causal, and not informational. One run and one conjecture. That is what a thesis is allowed to rest on before E1, and not after.

1.3 What the vista inherits from year one

Two inheritances, and the second is the one that decides whether the word *causal* can be used at all. The first is the failure vocabulary. FOE-Dreamer already names the two borrowed failures, objective mismatch and model exploitation, and already says why the adversarial case is not either of them: in both borrowed cases the drift is an artefact of the training loop, and here it is an opponent’s objective, so uniform predictive loss is selected against.

The second is the identifiability caveat, which the FOE-Dreamer page states about itself and does not soften.

“One assumption may not hold: the adversary latent may not be identifiable from defender-side observation at all.”

website, FOE-Dreamer, ongoing work

“The fallback is drift detection on the joint latent, which surrenders attribution and keeps the alarm.”

website, FOE-Dreamer, ongoing work

The foundations argument behind that sentence is a counting argument and it is short enough to carry here, because everything in this vista depends on it. Nonlinear independent component analysis is unidentifiable without extra structure, and what restores identifiability is an observed auxiliary variable that modulates the latent distribution. A world model of this family has two such variables available, the action and the reward. The action separates controllable from uncontrollable. The reward separates reward-relevant from reward-irrelevant. Two binary signals give four blocks, and the defender faces three sources of variation: defender effects, user activity, attacker activity. User activity and attacker activity land in the same block. Both are uncontrollable and both are reward-relevant, so neither signal can tell them apart.

“So with the action and the reward as the only auxiliary signals, u_t is identifiable as uncontrollable reward-relevant variation. It is not identifiable as user activity.”

FOE-Dreamer foundations, factoring

“the opponent model is not a component sitting beside the factoring. It is a precondition for the factoring’s interpretation”

FOE-Dreamer foundations, factoring

Of the four assumptions the argument lists, two hold and two fail, and both failures trace to one fact: the defender’s containment actions act on users as well as on attackers. Isolating a host drops user sessions, so exogeneity fails. The disruption penalty is a product of the defender action and the user state rather than a sum, so additive reward separability fails. That is the operational reality the whole line is about, and it is more interesting stated than omitted.

The consequence for this vista is direct. A third auxiliary signal buys the separation, and the only one on hand is privileged attacker supervision. Every causal claim this project wants to make about an operational network has to name the signal that licenses it, or it is a claim about one run.

2 Active

Terms I have to define on the spot, without looking anything up. Four fields each.

2.1 The causal words

Five objects have to stay apart, and the planning notes keep them apart in one table: predictive world model, causal representation, causal discovery, causal-mechanism learning, decision-relevant abstraction. Collapsing any two of them is how a project claims a causal result it did not produce.

causal world model

“a decision model that uses an explicit causal representation, causal graph, or causal mechanism to support prediction, intervention, counterfactual reasoning, transfer, explanation, or planning”

planning notes, Causal World Models, operational scope

Definition: A world model whose representation is committed to claims about what happens under intervention, and not only to claims about what comes next.

Distinguishes it from: A predictive world model, which is trained on likelihood and scored on continuation. Also from a structural causal model on its own: the planning notes hold that an SCM representation does not by itself constitute causal reinforcement learning, because causal reinforcement learning adds assumptions or prior knowledge in order to analyze the mechanisms connecting actions to consequences.

Operational test: Name one query the model answers that association cannot, and name the assumption that licenses the answer. If neither exists, the word is decoration.

In this project: The object the vista is named for, and the object whose necessity has to survive a comparison against four simpler alternatives before it is built.

causal representation

Definition: High-level causal variables extracted from low-level, entangled observations. The prior step: before anything can be said about edges, the nodes have to exist.

Distinguishes it from: Causal discovery, which assumes the variables are already given and recovers relations among them. A representation alone does not identify all directed mechanisms.

Operational test: Can you point at the observation channels that make up one variable, and say what would change if two of them were split into separate variables?

In this project: The FOE-Dreamer latent blocks are candidate causal variables and nothing more until an auxiliary signal separates them. This is the entry that the identifiability argument attaches to.

causal discovery

Definition: Recovery of the relations among supplied or learned variables from observational or interventional data.

Distinguishes it from: Fitting a sparse graph. Observational data generally does not uniquely identify a directed acyclic graph without additional assumptions, so what comes back is an equivalence class unless something extra paid for the orientation.

Operational test: For each reported edge, name the intervention or the functional assumption that oriented it. An edge with neither is an adjacency.

In this project: Recovery evaluation has to report adjacency, orientation, equivalence-class compatibility and prediction separately, because the method families give different guarantees while fitting the same observational distribution.

causal mechanism

Definition: The structural function describing how a variable's direct causes and its exogenous noise generate its value.

Distinguishes it from: The graph. A graph without the relevant mechanisms is insufficient for every interventional or counterfactual query, so a recovered structure is not yet a model you can query.

Operational test: Ask for a numeric answer to a counterfactual. If the object can only say which arrows exist, the mechanisms are missing.

In this project: The distinction decides which of my planned outputs is reachable. A drift detector needs a graph. A counterfactual account of why a containment action was taken needs mechanisms.

decision-relevant abstraction

Definition: A representation whose parts are selected by whether removing one would conflate states that require different actions.

Distinguishes it from: A causal representation. Decision relevance does not establish that a concept is causal, and causal identification does not establish that a concept is decision relevant. They are two criteria and the project needs both stated.

Operational test: Delete the candidate concept and check whether the optimal action changes anywhere in the state space. That is the whole test.

In this project: The cheapest competitor the causal component has to beat. If a selected non-causal abstraction of the same size does the same work, the evidence supports the simpler architecture and this vista shrinks by one adjective.

identifiability

“the question is whether the intended decomposition is recoverable from the data at all, or whether the architecture merely happened to land near it once”

FOE-Dreamer foundations, factoring

Definition: Whether the quantity a model claims to represent is recoverable from the data and

the assumptions at all, as against a quantity a fitted model happened to land near on one run.

Distinguishes it from: Accuracy. A model can predict well and be unidentified, and an unidentified latent can be relabelled without changing a single prediction. Also from interpretability, which is a property of how the object is presented rather than of what the data can support.

Operational test: Name the auxiliary signal or the intervention variation that separates the claimed factor from its neighbour. If none exists, the claim is about the run.

In this project: The joint between the three adjectives. Trustworthy attribution rests on identifiability, and identifiability rests on which interventions the environment allows, which is a property of the generator and the testbed rather than of the learner.

intervention target

“The generator will ideally support us to disturb the learning agent by manipulating the configurations that it enables us to control.”

brainstorm, 9 August, the experiments bullet

Definition: The factor a given configuration change actually moves. The set of them, written as a matrix over factors and configurations, is what determines whether distinct factors can be told apart.

Distinguishes it from: A perturbation applied to a trained model to test it. An intervention target is a property of how the data was collected, not of how the result was probed.

Operational test: For each planted factor, is there a configuration that moves it and leaves its neighbour still? CITRIS makes the negative case sharp: two factors cannot be uniquely identified when they are always intervened on jointly or never intervened on.

In this project: The one identifiability requirement that costs me nothing extra, because the Learn Structure generator already plants factors at known degree with ground truth recorded, at about a minute per experiment. The matrix has to be written down before E1, not after.

2.2 The trustworthy words

trustworthy

Definition: Stipulated narrowly, the way *operational* is stipulated on the FOE-Dreamer page. A model is trustworthy here when the errors it makes are ones the defender survives, and when it is wide where it is wrong. It is not a claim that the model is accurate, and it is not a report of how confident an operator feels.

Distinguishes it from: Predictive accuracy, which does not measure this at all. Also from calibration in the ordinary sense, which asks whether stated confidence matches observed frequency across the data distribution, while this asks whether the width rises in the particular region an adversary will choose.

Operational test: Two questions, and both have to answer yes. Does the model’s uncertainty rise where its error rises? And does an operator reading it as uncertain do something different from an operator reading it as certain? If the second is no, the width is not doing operational work.

In this project: The first adjective in the title, and the reason the two planned outputs are a failure taxonomy for factored world models and a drift detector rather than a better score.

security-relevant failure

“fundamental, focused, and insightful challenges, that necessitate our methods, but also highlight the perils should it be done wrong”

brainstorm, 9 August, the domain bullet

“When does model error become strategically dangerous rather than merely predictive noise?”

website, World Model Failure track, core question

Definition: A model error that an adversary can reach, that changes a defender action, and that has a consequence for an asset. Three conditions, and all three have to be shown.

Distinguishes it from: A large prediction error, which stays a technical limitation until reachability, action change and consequence are demonstrated. The classification rule the catalogue uses is the same one: a limitation becomes a vulnerability only when a threat source can exploit or trigger it, and a risk only when likelihood and adverse impact are addressed.

Operational test: Name the actor, the path, and the asset. Any of the three missing and the entry is a limitation, and the catalogue says so rather than inflating it.

In this project: Nine mechanisms are already classified this way and only one carries a system-weakness class, causal-model misspecification. The stated coverage gap is that no evidence yet demonstrates a cyber attacker exploiting a causal world model’s graph or mechanisms in an autonomous defender. That gap is the project, and writing it down as a gap is what keeps the project honest.

model bias

Definition: The share of failure attributable to the world model’s own error, read directly off open-loop prediction error at fixed horizons rather than inferred from return.

Distinguishes it from: Failure attributable to the replay buffer or to the actor and critic. The toggle plan separates all three with one-flag decision tests, which is what makes attribution a measurement instead of an argument.

Operational test: Open-loop error at horizons 1, 5 and 10, plus the imagination-horizon sweep at 5, 10 and 15, which exposes how far model error pollutes the value targets.

In this project: The null hypothesis’s only explanatory variable, and the reason the null is a real competitor rather than a rhetorical one. E2 measures it on the testbed, so the structural thesis has to beat a quantity that is already on the table.

drift

Definition: Movement of the opponent latent away from the region the model was fitted on, in a direction the opponent chose.

Distinguishes it from: The borrowed sense in model-based reinforcement learning, where a policy drifts into states the model never saw and profits from the error there. In that sense drift is an artefact of the training loop. Here it is somebody’s objective.

Operational test: Ask what produced the movement. If removing the adversary removes it, it was the loop.

In this project: The detector is one of the two planned outputs, and its success criterion is already fixed in advance: it fires before the attacker reaches the region the model is wrong about, at a false-alarm rate an operator will accept.

2.3 The controllable words

controllable latent

“The controllable latent z_t carries host states whose evolution is action-conditioned: isolation status, deception deployment, observable reboots, and the alert dynamics that follow defender intervention.”

FOE-Dreamer, approach

“The exogenous latent u_t carries user activity, service load, periodic maintenance, and other observation-affecting variation that neither the defender nor the attacker controls.”

FOE-Dreamer, approach

Definition: The block of the latent state whose evolution is conditioned on the defender’s own action. Its complement is exogenous, and the boundary between them is enforced by architecture, through separate posterior and transition heads, rather than by a penalty added afterwards.

Distinguishes it from: Controllable in the title’s sense, which is about a person steering a deployed system rather than about a partition of a latent. The two senses share nothing except the word, and Section 5 keeps them apart.

Operational test: The check the paper already states: u_t tracks users and ignores the defender, z_t tracks the defender and ignores the user. Linear decoding from each block, both directions, both reported.

In this project: Under the two available auxiliary signals the block is identifiable as uncontrollable reward-relevant variation and not as user activity. Any vista claim that names it user activity is spending a third signal, and has to say which one.

rule-like objects

“If we can learn some ‘rule like’ objects together with the defense policy, or the adversary, it will be really helpful.”

brainstorm, 9 August, the transparency bullet

Definition: An object learned alongside the policy that a person can read as a rule. Three candidates are on the table and they are different objects: a causal graph edge, a selected decision-relevant concept, and an extracted rule of the kind rule-extraction methods produce from a deployed model.

Distinguishes it from: A post-hoc explanation of a policy that was never trained to produce one. Both remain live, and the architecture table keeps policy-plus-extracted-rules as a full alternative rather than as a fallback.

Operational test: Would removing the object change a decision? If not it is commentary, however well it reads.

In this project: The three candidates need separate fidelity tests. None of them establishes a faithful causal account of a sequential decision on its own, and treating any one of them as if it did is the fastest way to overclaim here.

transparency for accountability

“for cybersecurity tasks there’s this dire need to fully understand the decisions made by AI. if not in real time, then in post-event analysis. Either way, we need transparency for accountability”

brainstorm, 9 August, the transparency bullet

Definition: The record a defender’s decisions leave, sufficient for a person to reconstruct why an action was taken and to attribute the outcome to something. Real time and post-event are two different products of the same requirement, and my own sentence already separates them.

Distinguishes it from: Interpretability of the model, which is a property of the representation and can be satisfied without any decision record existing. Also from a fluent rationale, which is an output rather than evidence.

Operational test: Hand an analyst the record and a planted bad decision. Can they find it, and how long does it take? Accuracy and time, not preference.

In this project: The systematic review found no per-action explainability in the surveyed CAGE Challenge 1 and 2 agent implementations, which makes this an unmet deployment requirement rather than a finishing touch. It is also the one requirement that separates the real-time use of causal information from the post-event use, and counterfactual reasoning is expensive enough in high dimensions that the two uses may not both be affordable.

dependence type

Definition: The mechanism by which one player’s or one device’s behaviour reaches another’s best action, in the five-way classification Learn Structure fixes: time, space, control, cause, information.

Distinguishes it from: The four contrast cases on the Learn Structure page, which includes the causal graph. Those are properties of the game as written. Dependence type is a property of how the game is played, and the whole classification exists because holding a game fixed and varying only the opponent’s behaviour leaves all four contrast cases unchanged while changing the reasoning the defender needs.

Operational test: Plant one type at a known degree and vary nothing else. The generator does this, which is the only reason the type profile is measurable at all.

In this project: The independent variable of E1, and the reason this vista sits on the Learn Structure generator rather than on the testbed alone. It is also where the thesis is most exposed: if the failure profile across types is flat at matched data budgets, the structural half of the thesis is not supported and the work becomes a horizon-and-bias analysis.

3 Working

Terms I have to recognise exactly. Looking them up is fine.

structural causal model

Definition: Endogenous variables generated by structural equations from their parents and from independent exogenous variables. A partially observable decision process can be written this way by treating observations, states, actions and rewards as endogenous.

Distinguishes it from: Causal reinforcement learning, which the survey treats as the addition of assumptions or prior knowledge on top of the representation, not as the representation itself.

Markov equivalence class

Definition: The set of graphs implying the same conditional independences, which is what constraint-based discovery returns from observational data.

Distinguishes it from: A recovered graph. Reporting one member of the class as though it were the answer is the most common way a discovery result is overstated.

faithfulness

Definition: The assumption that every conditional independence in the distribution is implied by the graph structure rather than arising from cancelling paths. Constraint-based correctness rests on it together with the Markov condition.

Distinguishes it from: A finite-sample guarantee. Pointwise consistency under these assumptions gives no error probability and no interval on the recovered structure, which is a real limit when the corpus is one testbed.

latent confounding

Definition: An unobserved common cause of two represented variables. Some methods assume it away and others are built for it, and the two return different objects.

Distinguishes it from: Partial observability of the state, which is about what the agent sees at decision time. Confounding is about what the estimator never sees at all.

continuous score-based search

Definition: The reformulation that replaces combinatorial acyclicity with a smooth equality constraint over a weighted adjacency matrix, so ordinary constrained optimisation applies.

Distinguishes it from: A solution to the search problem. The equivalent continuous program is still nonconvex, so the reformulation moves the difficulty rather than removing it.

functional causal model

Definition: An effect modelled as a function of its direct causes plus independent noise, with restrictions on the function and noise classes strong enough to break ties inside one equivalence class.

Distinguishes it from: An intervention. Both can orient an edge and only one of them is free.

objective mismatch

Definition: The model minimises prediction error across the data distribution while the policy needs accuracy only where it will act. One-step likelihood can improve without control improving, and a globally accurate model can be locally wrong exactly where the decision is.

Distinguishes it from: Model exploitation, where a policy drifts into states the model never saw and profits from the error there. Both are named in the FOE-Dreamer page and both are artefacts of the loop.

imagination horizon

Definition: How many steps a policy is optimised over inside the model. Model error compounds with it, which is why the conservative line in this literature branches short rollouts from real data.

Distinguishes it from: Episode length. The horizon is a training choice and the episode is the environment's.

threat model

Definition: An explicit record of protected assets, system boundaries, adversary goals, knowledge and capabilities, reachable attack-surface elements, exploitable weaknesses, adverse events, impacts, likelihood assumptions and controls.

Distinguishes it from: A list of all model failures, since a vulnerability needs exploitability and a risk needs likelihood and impact. Also from a red-team result, since the model states the testable assumptions before an exercise runs them.

attack surface

Definition: The boundary points where an attacker can enter, cause an effect, or extract data. For a learned defender: observation channels, training data, world-model state, planner interfaces, tools, permissions, logs and analyst-facing outputs, and only those an attacker can actually reach.

Distinguishes it from: The component inventory. Listing a module is not naming a reachable boundary point.

vulnerability

Definition: A weakness in a system, procedure, control or implementation that a threat source could exploit or trigger.

Distinguishes it from: A prediction error, which becomes one only when a threat source can exploit or trigger it in the deployed system. This is the single distinction my brainstorm asked for by name.

risk

Definition: The extent to which an entity is threatened, as a function of adverse impact and likelihood.

Distinguishes it from: A vulnerability, which needs no likelihood statement.

red teaming

Definition: Authorised adversarial testing that searches for any way to defeat the system.

Distinguishes it from: Adversary emulation, the narrower case that fixes an adversary profile and behaviour chain from documented techniques and tests against that. The narrowing is the whole difference, and it is the one I was using loosely.

LLM-only defender

Definition: An architecture in which the language model interprets observations, plans, and selects actions, with no learned state model beside it.

Distinguishes it from: The three architectures that add one component each: a non-causal state tracker, a predictive world model, a causal world model. Plus the fifth, a learned policy with post-hoc extracted rules.

explanation fidelity

Definition: Whether an explanation identifies the evidence and the mechanism behind an action, as against whether it reads well.

Distinguishes it from: Analyst satisfaction. The four candidate explanation objects, causal graphs, selected concepts, extracted rules and model rationales, each need this tested separately.

4 Peripheral

Enough to place them and retrieve them when needed.

- **Transportability and external validity.** The formal question of when a result carries from the population it was estimated on to another. Sits directly behind sim2sim in the Toward Deployment cluster and is on my reading list already.
- **Nonlinear independent component analysis.** The unidentifiability result the whole factoring argument descends from, and the source of the auxiliary-variable requirement.
- **Denoised MDP and the factored-latent line.** The two-by-two of controllable against reward-relevant, which is not a taxonomy chosen for tidiness. It is the finest partition two binary signals can support.
- **Adversarial policies.** A strategic opponent exploiting a policy with natural observations outside the victim's training distribution, without being a generally stronger opponent. The cleanest existing demonstration that exploitation is not the same thing as strength.
- **Component integrity against system availability.** The argument that loss of an intrusion model's integrity can affect the availability of the controller and its managed network, which is the pathway a model-level error needs before it counts as a security consequence.
- **The evaluation numbers from the domain review.** Partial observability unaddressed in 60.6% of 66 reviewed papers, oversimplified environments in 40.9%, missing convergence evidence in 71.2%, missing variance analysis in 66.7%. And the adaptive-opponent case: a defender trained against one attacker scoring -19.0 against it and -104.1 against another.

5 Where the words disagree

Four places where two of my own files use one of these words in two ways. None of them is resolved here. Three of them I would resolve one way today and differently next month, which is the reason to write them down instead.

Controllable, two senses that share nothing

Inside FOE-Dreamer, *controllable* names a block of the latent state, the one whose evolution is conditioned on the defender's action. Its complement is exogenous. The word is doing partition work and its opposite is *uncontrollable*.

In the title of this vista, *controllable* names a property of the deployed system: an operator can constrain what the defender does, can stop it, and can say afterwards why it acted. The word is doing governance work and its opposite is something closer to autonomous.

The two do not overlap. A model can factor its latent perfectly and be ungovernable, and a fully governed defender can carry a single undifferentiated latent. Any sentence in this vista that uses the word has to say which one, and the safe choice is to keep *controllable latent* as one headword and never let the bare adjective stand alone.

Causal, four senses, two of them in the same file

On the Cyber World Modeling page the word describes the environment: the network's own dynamics are close to stationary and roughly causal, meaning reachability, topology and load change by mechanism rather than by strategy.

On the Learn Structure page the causal graph is named as one of four contrast cases, properties of the game *as written*, and the page's whole point is that strategic dependence is not any of them.

On the same Learn Structure page, *Cause* is one of the five mechanisms in the dependence table, described as what one player's actions change that the other depends on. It is also the type the generator plants and the type the learner errs on most.

In this vista, causal is the property the representation is supposed to have.

Four senses, and the second and third are on one page. They are not contradictions, but they are not one concept either, and a reader who takes the vista's sense back to the Learn Structure page will read the contrast list as a disavowal of the vista. It is not. The contrast list is about the graph of the game, and the vista is about the graph of the dynamics.

Many models, or one world model

The site nav and the page heading say *Many Trustworthy, Controllable, Causal Models*. The folder slug says `trustworthy-controllable-causal-world-model`, singular, with *world*. The collection page says *trustworthy, controllable, causal world models*, plural, and calls it a manuscript line.

The plural is not a typing slip and it is not cosmetic. Many models is a composition problem, and it is one of the two Cyber World Modeling questions that no page in that cluster develops:

“Given a set of world model component, how can we compose the most “suitable” model for the target operational environments?”

website, Cyber World Modeling, ongoing and future work

That question survives in exactly one place on the site, and it is this title. Deciding whether the vista is about one causal model or about composing many is therefore not a naming decision. It decides whether the fourth open question of the year-one cluster has a home.

Trustworthy, of a model or of a result

On the Daedalus page the word describes a result: most autonomous-defence work happens in a simulator, and the results are only as trustworthy as the simulator's fidelity. That is an epistemic property of a study, and the thing that fixes it is realism of the environment.

In this title the word describes a model in operation, and the thing that fixes it is honest width where the model is wrong.

The two senses pull in different directions when they meet. A model that widens its uncertainty makes for a less quotable result, and a testbed built to make results trustworthy in the first sense is expensive enough that it constrains how many configurations of the second sense can be measured. Both are already in play here, since the operational testbed and the controlled generator are the two evidence sources of this vista and each serves one sense.

Perils, dangerous, security-relevant

I wrote *perils*. The restatement layer wrote *make failures of those methods dangerous*. The planning notes write *security-relevant* throughout, with a classification rule attached to it. My own track page asks when model error becomes *strategically dangerous rather than merely predictive noise*.

Only one of the four carries a test. Security-relevant has three conditions and a classification table behind it. The other three are the same idea before it was made checkable, and the reason to record them together is that the checkable version is narrower than what I meant, and I should know that when the catalogue comes back and classifies most of my worries as ordinary technical limitations.

Pure, and what got substituted for it

I flagged this one myself, mid-sentence, in my own note:

"pure (how pure can it be, i don't know, it's a vague term) llm-based defender agents"

brainstorm, 9 August, the integration bullet

The planning notes substitute *LLM-only defender* and give it a definition: the language model interprets observations, plans, and selects actions. That is a clean object and it is testable, which is more than my word had. It is also narrower than what I was worried about, since an agent with retrieval, tools and memory is not pure by any reading and is what industry actually deploys. The substitution is carried in the entry above, and this box is the record that it was a substitution.

6 What this vista has to settle

Four things, in the order that each one blocks the next.

The intervention-target matrix. Which generator configuration moves which planted factor, and which pairs only ever move together. Until that is written down, no result from the generator can be called an identification, and every causal word in the title is on loan. This is cheap and it is

not done.

Whether the type profile exists. E1 is the only experiment that separates the thesis from the null, and the decision rule is already declared: a flat profile across types at matched data budgets means the structural half is not supported and the work becomes a horizon-and-bias analysis. Declaring that in advance is the part I want to keep, because it makes the day-seven decision a read-off rather than a judgement call under a deadline.

Which causal information is usable in real time. Counterfactual reasoning is expensive in high-dimensional tasks, and post-event analysis and real-time control are different uses of the same object. My own sentence already separates them and my own plans keep treating them as one deliverable. They are two, and they may not both be affordable.

Whether a simpler alternative supplies the same benefit. The decision rule for the whole project is already written and it can end the project: causal integration is supported only if it resolves a documented security-relevant failure more reliably than the LLM-only, state-tracking, predictive-model and rule-extraction alternatives. If state tracking or rule extraction supplies the same benefit, the evidence supports that simpler architecture. And if nothing succeeds, the result identifies an unresolved problem without proving that causal modeling was the missing piece.

That last one is the honest shape of a five-year vista. Three adjectives in a title, and a written rule that can take two of them away.

“If there’s a genie of world modeling, what wishes should I make?”

website, Headspace

“When does an agent have the need to model the world in the first place?”

website, Headspace

Three wishes are already in the title. The second question is still the better one.

7 Sources

- My own words: `Notes/08-09-brainstorm-project-planning.txt` in the world-model-failure folder, for the vocabulary request, the perils sentence, the generator sentence, the transparency sentence, the rule-like objects and the parenthetical about *pure*. Everything marked *brainstorm* is from that file.
- My own pages: the vista page and the 5-Year Vista index; the LucidWorld collection page and the World Model Failure track page for the core question and the four buckets; the five Cyber World Modeling pages for the two challenges, the identifiability caveat, the failure vocabulary, the honest-against-silent preference and the composition question; the Learn Structure page for the five dependence mechanisms and the four contrast cases; the Daedalus page for the second sense of trustworthy; Headspace for the genie question.
- My own submission plan: `Latex/sections/usenix_one_two_pager.tex` and `Latex/sections/three_week_plan.tex` in the vista folder, for the thesis, the null, E1 to E3, the two contributions and the declared fallback, with `reference.bib` for what each claim is sourced to.
- The planning notes, written against the brainstorm and cited back to it:

`causal_world_models.tex` for the operational scope and the five-way distinction; `causal_structure_learning.tex` for the method families, the conditional identifiability section and the CITRIS condition; `network_defense_challenges.tex` for the domain evidence and the percentages; `agentic_defender_integration.tex` for the five architectures and the decision rule; `threat_modeling_red_teaming.tex` for the security vocabulary and the red teaming against adversary emulation narrowing; `cybersecurity_challenge_catalogue.tex` for the classification rule and the nine mechanisms.

- The FOE-Dreamer foundations file `f01_factoring.tex` for the auxiliary-variable argument, the two-by-two, the three sources of variation and the four assumptions of which two fail. The approach section of the paper for the two latent blocks and the architectural factor boundary. The toggle plan for open-loop error at horizons 1, 5 and 10 and the imagination sweep.
- Shared cluster terms, world model, factoring, opponent latent, operational, drift and the empirical game, are defined once in `cwm_shared.tex` and are carried here only where this vista fixes something extra about them.